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RESEARCH ARTICLE

Dynamic Automated Labeling System for Real-Time User Intention Analysis

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ABSTRACT Real-time online user responses can be useful for identifying people's reactions to business strategies or social phenomena. Many approaches have been proposed to classify user responses, yet most of them exploit either manual or automatic labeling that fails to reflect real-time information, preventing use as a generalized model. In this study, we propose a dynamic automated labeling technique that considers changes in user response over time by combining LDA (Latent Dirichlet Allocation) topic modeling and word2vec. Topic modeling is used to extract topics that will be used as true labels, and word2vec is used to map the dominant topic into label to annotate the users' textual data by calculating the vector similarity. Experimental results on disaster-related Tweet datasets show that our approach outperforms other baseline approaches of automated annotation in terms of classification performance. The technique presented in this study can be applied in business to create an automated system to analyze and continuously monitor users' real-time reactions.

INDEX TERMS Dynamic automated labeling, automatic text annotation, real-time user analysis, topic modeling, text classification, text mining.

I. INTRODUCTION

Online user-generated text data can play an important role in understanding people's opinions and emotions related to any event, product, service, or idea. Huge amounts of user response data are generated daily on microblogging websites such as Twitter, and an automatic analysis of such real-time text data can provide meaningful information on users, consumers, and businesses [7]. Particularly during the onset of a crisis, people use social media platforms to post situational updates, look for useful information, and ask for

help [13]. Such communication can also help humanitarian organizations gain situational awareness about urgent needs of affected people, critical infrastructure damage, and medical emergencies [17]. Although user reviews in the business domain are not as plentiful as messages posted on microblogging platforms, they can be useful resources for business stakeholders to understand the latest customer reactions to business activities such as marketing or service updates. Mobile app user reviews on Google Play Store or Apple's Appstore provide information on user experiences [19], bugs [9], requests for new features [11], and user evaluations [15] and can be utilized for mobile service improvement in business.

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The main objective of text classification for both disaster- and business-related text data is to identify the topic of a user response. For text classification problems, it is essential to generate data labels. However, current text classification models suffer from lack of real-time-labeled data, hindering use as generalized models [4], [15]. Traditional manual labeling is both time-consuming and expensive and could affect classification performance due to incorrect labeling caused by researcher error or subjectivity [2], [8], [16]. In addition, data annotation relying on human intervention can quickly become infeasible when dealing with a large amount of information [3], such as real-time Tweets [1].

Although many studies about automated annotation have been conducted, compared to generative models, classifier-based methods do not accommodate new, previously unseen topics. These methods are a poor fit for processing dynamic data volumes of, for instance, online customer reviews [10]. Therefore, we present a dynamic automated labeling technique where labels are continuously updated based on changes in user response over time. The contributions of this study are as follows.

- (1) By combining LDA (Latent Dirichlet Allocation) topic modeling and word embedding, we propose a dynamic automated labeling technique in which changes in social trends or user response over time are continuously reflected in labeling, instead of using fixed labels.
- (2) The dynamic automated labeling technique proposed in this study can be extensively applied to text data in other domains such as business, society, politics, e-commerce, and medicine.
- (3) In comparison with other baseline approaches, we evaluate our framework from the perspective of classification measures and reached significant results.
- (4) The technique proposed in this study can be utilized to build an automated labeling and classification system for real-time user response analysis in the business domain.

In this study, we propose an approach to automatically annotate and classify text data using LDA topic modeling and word2vec. First, LDA topic modeling is used to derive underlying topics from documents and keywords representing each topic. The documents then are converted into feature vectors by word2vec and are assigned to the dominant topic based on vector similarity between the documents and the derived topic keywords. With the newly annotated data through our proposed labeling approach, we train deep learning models and evaluate the classification performances. Although the datasets used in the experiments in this study are limited to the field of business (Korean stock trading app reviews) and social issues (disaster-related Tweets), the proposed framework can be extended to all types of text data.

II. LITERATURE REVIEW

A. STUDIES ON USER RESPONSE ANALYSIS

In the study of Maalej et al. [16], app reviews collected from 1,186 popular apps on the Google Play Store and the Apple's Appstore were manually labeled and classified into four categories – bug reports, feature requests, user experiences, and star ratings. Text mining techniques of BoW (Bag of Words), stopword processing, lemmatization, metadata, and sentiment analysis were applied to the dataset for comparison purposes. Classification performance was evaluated based on the combination of these techniques.

Guzman et al. [8] presented a classification model that classifies app reviews into seven categories in relation to software evolution: bug reports, app feature strength, app feature weakness, user demand, praise, complaints, and user scenarios. The following seven apps were selected for the analysis: Angrybirds, Dropbox, Evernote, Tripadvisor, PicasArt, Pinterest, and WhatsApp. The collected reviews were manually labeled in one or more categories, and multi-label classification models were applied to train classifiers for each category.

Aslam et al. [2] also adopted manual labeling for app reviews collected from 1,180 popular apps and classified the reviews into four categories – bug reports, enhancement reports, user experiences, and ratings. In this study, non-textual information, such as app size and number of app installations, as well as sentiments extracted from the reviews were combined with texts and were used to train the classifiers.

Caragea et al. [4] presented a CNN (Convolutional Neural Network)-based classification model that classifies six flood-related Tweets into two categories: informative and not informative. The study used an already labeled dataset along with BoW and n-grams used as features for model training.

In a study by Li et al. [15], a domain adaptation approach was used to classify six Tweets related to hurricane, flood, or bombing events into two categories: relevant and irrelevant. A Naïve Bayes classification model was presented. The study used an already labeled dataset; after training with only labeled prior source disaster data, the labels of predicted target disaster data were predicted. The target disaster data were classified with high reliability (top k based on posterior probability) and were used for learning in the next iteration to predict new target disaster data.

B. STUDIES ON AUTOMATED LABELING

Yang et al. [25] proposed an automatic labeling method for Korean game app reviews with eight categories – payment, account, server, structure, interface, character, system, and others – using word2vec and the concept of vector similarity. Final labels were assigned to the reviews based on the extended jacquard similarity between words that appeared in the reviews and representative keywords for each category. In addition, Yang et al. [25] conducted an experiment to determine threshold of similarity score differences for

multi-labeling, and they selected the final threshold that best improved the model performance.

Tang et al. [20] presented a BERT and word2vec-based automated customer complaint labeling model for hierarchical classification. Text enhancement was applied due to imbalance of the dataset. Based on the hierarchical structure of customer complaints, shallow labels, which are the second hierarchical level, and deep labels, which are third hierarchical level, were finalized.

Hananto et al. [10] proposed a segmentation-enhanced topic modeling approach for automated text annotation. LDA topic modeling is used to derive the optimal topic categories from online reviews, and the TopicDiff-LDA algorithm is proposed to divide reviews into semantically homogeneous segments for multi-topic reviews. The segments are used to update the topic model by recalculating the distribution probability of derived topics with a set number of topic categories.

Tushev et al. [21] presented keyATM, a keyword-assisted approach for generating topic models, to extract domain-specific information from mobile app reviews in the domains of investing and food delivery. KeyATM can generate more cohesive topics than the LDA model using seeding keywords extracted from the review corpus.

Karami et al. [14] presented a Situational Analysis framework using a dataset of flood damage Tweets. The Tweets were automatically labeled as positive, negative, or neutral based on the LIWC (Linguistic Inquiry and Word Count) lexicon, and negative Tweets were used to derive crisis-related situational information by LDA topic modeling.

Gupta and Joshi [7] presented an automatic labeling technique based on k-means cluster analysis for sentiment classification of Twitter datasets related to lockdown, an Indian lights-out event known as 9pm-9minute, and demonetization. Feature engineering techniques such as POS (Part-of-Speech)-based, n-grams, lexicon-based, and negation handling were applied to improve clustering performance. These features were used as input for k-means clustering, and the three final clusters of positive, negative, and neutral were assigned as labels to the Twitter dataset.

Wahid et al. [22] proposed a method of automatically labeling crisis-related Twitter data by combining LDA topic modeling, ranking algorithm, and BERT embedding. The dominant topic was selected based on a ranking algorithm called TF-IDF (Topic Term Frequency and Inverse Document Frequency) and was assigned to each Tweet as a final label.

Fig. 1 is a comparison table of previous studies regarding customer reviews based on the data used in the study, labeling techniques, classification modeling, and applications. For novelty of labeling, studies that adopted manual labeling were classified as 'insufficient,' and studies that applied automatic labeling were classified either as 'good' or 'excellent.'

III. BACKGROUND

A. TOKENIZATION

In natural language processing (NLP), each corpus consists of multiple tokens to be used as input for NLP models, and

tokenization is a process of splitting corpus into tokens. Word tokenization can be performed based on whitespace for languages like English, but it cannot be applied to agglutinative languages like Korean, where words consist of a sequence of multiple morphemes. Thus, morphological analysis should be used for word tokenization in natural language processing of Korean.

B. EMBEDDING

Embedding in NLP refers to a technique where words or documents are represented as numeric vectors. The embedding vectors contain semantic information of documents that enable the similarity calculation between words or documents and can be used as input for deep learning models. Thus, choosing an appropriate embedding approach is important in model performance.

Embedding techniques can be divided into frequency-based embedding, matrix factorization-based embedding, prediction-based embedding, and topic-based embedding, depending on how documents are represented. Frequency-based embedding techniques, such as BoW, TF-IDF, and count vector, simply represent documents based on word frequency. Matrix factorization-based embedding, which utilizes SVD (Singular Value Decomposition) on a DTM (Document-Term Matrix), includes LSA, GloVe, and Swivel. Prediction-based embedding is used to predict an upcoming or nearby words given the surrounding context and includes techniques like word2vec, doc2vec, FastText, BERT, ELMo, and GPT. Topic-based embedding, such as LDA, is used to infer latent topics from documents.

Documents can be converted to embedding vectors based on either a word or sentence. Word2vec, FastText, LSA, GloVe, and Swivel are word embedding techniques, whereas LDA, doc2vec, ELMo, GPT, and BERT are sentence embedding techniques. Since word embedding techniques capture the semantic information of words into embedding vectors, it is hard to distinguish homonyms using word embedding. However, sentence embedding captures the semantic information of word sequences (rather than individual words), it shows better performance with transfer learning.

C. WORD EMBEDDING: WORD2VEC

The word2vec technique includes a Continuous Bag of Words (CBOW) that predicts a central word based on the surrounding words and a Skip-Gram that predicts the surrounding words based on a central word. That is, the context and distribution information around a specific target word are implied in the embedding, and a word having a similar meaning can be found by calculating the similarity between the transformed vectors. The two structures of word2vec are shown in Fig. 2.

When a document is converted into a vector by word2vec, the similarity between two documents can be measured by calculating the cosine similarity using the angle formed by the vectors rather than the magnitudes. If the angle of the two vectors is 0° , the cosine similarity is 1, and the two documents are the same. If the angle of the two vectors is 90° , the cosine

Excellent

Good

Average

Insufficient

N/A

Previous studies	Dataset		Labeling			Classification modeling		Research application	
	Size	Diversity of data sources	Novelty	Reliability	Effectiveness of the labels	Performance (F1-score)	Additional methods for performance improvement	Application to different domains	Application to business
Maalej et al. (2015)	Average	Good	Insufficient	Average	Average	Average	Good	Insufficient	Insufficient
Guzman et al. (2015)	Average	Average	Insufficient	Average	Average	Insufficient	Average	Insufficient	Insufficient
Aslam et al. (2020)	Average	Good	Insufficient	Average	Average	Excellent	Good	Insufficient	Insufficient
Yang et al. (2019)	Good	Average	Average	Average	Average	N/A	Good	Insufficient	Average
Tang et al. (2021)	Insufficient	N/A	Excellent	Good	Good	Insufficient	Average	Average	Average
Hananto et al. (2022)	Insufficient	N/A	Good	Excellent	Good	Good	Good	Average	Average
Tushev et al. (2022)	Excellent	Average	Average	Excellent	Excellent	N/A	Average	Insufficient	Average
My approach	Good	Average	Good	Excellent	Good	Good	Good	Good	Excellent

FIGURE 1. Previous research on user response analysis.

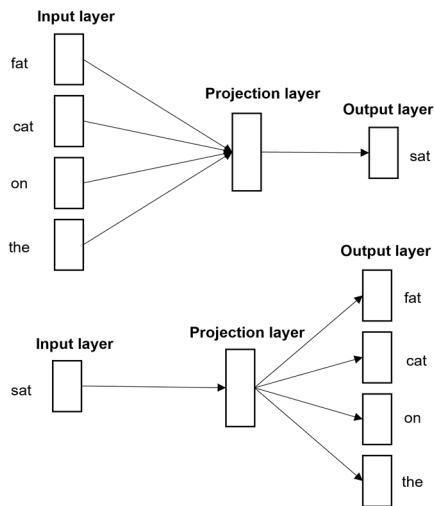


FIGURE 2. Structures of CBOW (upper) and Skip-gram (lower).

similarity is 0, and the two documents are not at all similar. Equation (1) shows the cosine similarity formula between two vectors X and Y , where $\langle X, Y \rangle$ is the dot product of the two vectors and $\|X\|$ is the length (norm) of vector X .

$$\cos\theta = \frac{\langle X, Y \rangle}{\|X\| * \|Y\|} \tag{1}$$

D. SENTENCE EMBEDDING: LDA

Latent Dirichlet Allocation, also known as Topic modeling, is a text mining technique that discovers underlying topics in documents. It is often used for post-analysis of a given text rather than for predicting a new document. Fig. 3 shows the structure of the LDA algorithm.

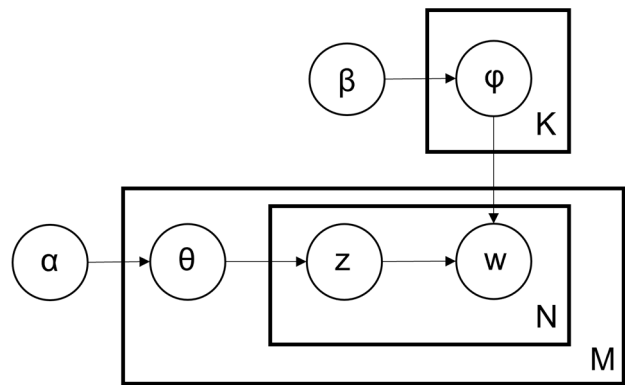


FIGURE 3. Structure of LDA topic modeling.

LDA uses the basic assumption that there are several topics composing documents, and that a set of words comprises each topic. The first purpose of topic modeling is to estimate the probability distribution of topics in the documents. It is assumed that each topic in the document has a probability value, and the probability distribution of each topic follows the Dirichlet distribution. In Fig. 3, M is the number of documents. θ is the topic distribution or Dirichlet distribution of the documents, and α is a parameter of the Dirichlet distribution.

The second purpose of topic modeling is to determine the distribution of words in each topic. Since each topic is composed of a set of words, the topic can be expressed by the weight of each word. In Fig. 3, K represents the total number of topics, and φ is the distribution of words in each topic and follows the same Dirichlet distribution. When the topic

to which word W in the document belongs is Z , N represents the frequency of words used in the document. LDA assumes that the word distribution of documents is determined by combining the topic distribution of the documents and the word distribution of each topic. That is, topic modeling is a technique for estimating the topic distribution of documents and the word distribution of each topic by measuring the frequency of words used in the document.

To select the appropriate number of topics K , the performance of topic modeling is measured by the perplexity and topic coherence. Perplexity is a measure of how similarly the Dirichlet model can generate a given document, where a smaller value indicates better performance. Topic coherence is a measure of whether the words occupying the top weight of each topic are semantically similar. Here, a higher value indicates better performance.

E. MULTI-CLASS CLASSIFIERS

1) RNN-BASED MODELS

The RNN (Recurrent Neural Network) is a deep learning model in which the output of the previous step affects the next step in the training process. Because it utilizes information from the previous step, it can be mainly used for processing sequential data, such as natural language. However, RNN suffers long-term dependency as the number of timesteps increases, because information from the earlier steps is not properly transmitted to later steps. That is, as the length of the document increases, the information from the front side gradually is lost and is not included in the training, possibly decreasing the prediction accuracy.

The LSTM (Long Short-Term Memory) technique has been proposed as an alternative to solve this problem. The input gate of LSTM receives the hidden state of the previous time and the input value of the current time and controls the amount of information to be stored in the current state. The forget gate determines the amount of information to be deleted among the hidden state from the previous timestep and the input value at the current timestep. Finally, the output gate determines the output value of the cell through the activation function.

On the other hand, the GRU (Gated Recurrent Unit) is an RNN-based model that solves the long-term dependency problem in a manner similar to LSTM, but it has a simpler structure. Unlike LSTM, which has input, forget, and output gates, GRU only has an update gate and reset gate to properly combine present and past information and transfer it to the next cell.

Lastly, BiLSTM (Bidirectional LSTM) utilizes two independent LSTMs, one from each side, enabling successful capture of the information in the later steps. Fig. 4 shows the structures of LSTM and GRU.

2) CNN

The CNN (Convolutional Neural Network) is a neural network that shows good performance in image processing and consists of a convolutional layer and a pooling layer. In the

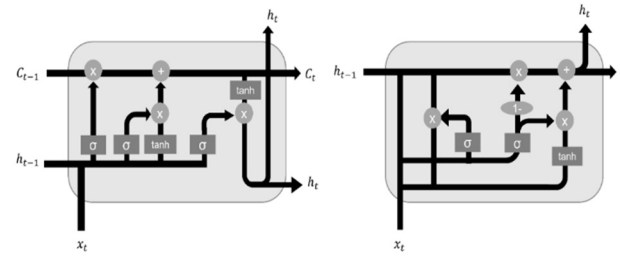


FIGURE 4. Structures of LSTM (left) and GRU (right).

convolutional layer, features of an image are extracted in the form of a feature map, and convolutional operations are performed while scanning the entire image with a matrix called a kernel (or filter). Elements of the kernel matrix are weights in the artificial neural network, and after weight calculation, they pass through an activation function such as ReLU to a pooling layer to reduce the size of the feature map. Max pooling and average pooling are mainly used for pooling. When CNN is applied to text classification, each sentence passes through an embedding layer and is processed in a 1D convolution operation as an embedding vector. In NLP, the size of a kernel is the number of words referred to in each operation.

3) BERT

The BERT (Bidirectional Encoder Representations from Transformers) approach is a method of pre-training language representation published by Google. Although it is based on a transformer that uses an encoder-decoder format, BERT only uses an encoder to perform word embedding given the context of words. The BERT model is pre-trained with many documents, such as Wikipedia, and can be used for various downstream tasks, including text classification, named entity recognition, and QA.

BERT's language processing performance for the Korean language is limited due to the small proportion of Korean in the training data. KoBERT (Korean BERT pre-trained case) was developed by the SKTBrain team to overcome these limitations. Based on 5 million sentences and 54 million words taken from Korean Wikipedia, the tokenizer was trained using SentencePiece, and the total number of vocabularies was 6,002. Since SentencePiece performs word separation without prior tokenization, it can be applied to languages that are difficult to tokenize, such as Korean.

IV. PROPOSED FRAMEWORK

A. DYNAMIC AUTOMATED LABELING

The dynamic automated labeling presented in this study is identical to general automated labeling techniques in the sense that they automatically assign labels to unlabeled data. However, in the case of automatic labeling, the type or number of labels is usually fixed in advance, possibly leading to unreliable classification results when those labels are applied

to previously unseen datasets that cannot be explained by existing labels.

On the other hand, the dynamic automated labeling proposed in this study is suitable for labeling previously unseen documents since it constantly reflects recent trends in labeling rather than simply using fixed labels. It is particularly appropriate for analyzing users' real-time responses in domains that undergo a rapid amount of change. Therefore, in this study, experiments were conducted on two kinds of datasets: a business dataset that contains Korean stock trading app reviews and a social dataset that contains disaster-related Tweets. These datasets reflect changes in trend or user responses over time and are unlabeled, making them suitable for the dynamic automated annotation.

One of the contributions of developing the dynamic automated labeling technique in this study is to build an automated system for monitoring and analyzing users' real-time intentions and reactions in business. Fig. 5 shows the overall process of establishing an automated system.

First, various text data that can identify user reactions are collected through web crawling or from external sources. These scattered data are stored in an integrated space, as cloud storage, and then are moved to an internal database and a big data platform to generate data marts for analysis. Afterward, dynamic auto-labeling is performed on the collected data, with labels constantly updated by trend analysis. Classification of new unseen datasets is performed based on the automatically assigned labels. Eventually, the classification results can be used to establish the automatic monitoring and analysis system for business stakeholders in service planning, development, and marketing. Of the entire process illustrated in Fig. 5, experiments conducted in our study are limited to web crawling, automatic labeling, and classification modeling due to resource constraints.

B. PROPOSED METHOD OVERVIEW

In this study, we first apply the automated labeling technique by incorporating LDA topic modeling and word2vec and then use deep learning algorithms for text classification. Fig. 6 illustrates the overall process of the proposed framework of this study, and this section will further discuss each step in detail based on the business dataset, which is Korean stock trading app reviews.

First, we collect user reviews of stock trading apps through web crawling. Afterward, text preprocessing and morphological analysis (which is equivalent to word tokenizing in Korean) are performed on the collected reviews.

Next, we adopt LDA topic modeling to generate topics and representative keywords constituting each topic. These keywords are later utilized to calculate vector similarities with the words (tokens) appearing in the original text.

The original text data are transformed into feature representation through word2vec, and a "Keyword-Word Similarity Matrix" is generated by calculating the vector similarities between the words used in the text data and the keywords extracted from the LDA topic modeling. After generating a

"Review-Word Matrix" indicating the frequency of words for each review, a "Review-Keyword Score Matrix" is constructed based on the dot product of the two matrices. By comparing the sum of the similarity scores of the representative keywords for each topic, the topic with the highest similarity score is determined to be the "dominant topic" for a review and is assigned as the final label. The given label is regarded as the true answer for the review and is used to train classification models.

We used RNN-based algorithms such as LSTM, GRU, and Bi-LSTM because they perform very well with sequential information like textual data. In addition, CNN and KoBERT also were adopted for classification modeling for comparison purposes. The model performance is evaluated based on the precision, recall, and F1-score. Also, the effectiveness of the proposed framework is verified by comparing the classification performance with those of seven baseline models related to automatic text labeling.

C. DATASETS

For this study, we collected user reviews of the 24 stock trading apps from 14 Korean securities companies provided by the Google Play Store using the BeautifulSoup and Selenium packages of Python. We chose the 24 stock trading apps based on the brand awareness of the securities companies in Korea, and popularity of the apps.

The total number of collected reviews was 41,705, and the collected items consist of the app name, user ID, review date, review content, and rating. If reviews are collected in the most recent order, it may include biased opinions. Therefore, a similar number of reviews was collected for each rating based on the five ratings left by users to reflect various opinions from users. Due to the nature of the Google Play Store website, the maximum number of reviews that can be collected per app was 3,200 (up to 640 per rating), but for apps with a small number of reviews, all reviews were collected. Table 1 shows the number of reviews collected for each app.

D. TOKENIZING

Fig. 7 illustrates the data preprocessing process of this study. Among the collected app reviews, 40,029 with five or more characters (including spaces) were selected for morphological analysis. Characters consisting only of special characters, English letters, numbers, and Korean consonants were considered stop words and were removed. After correcting typos, review data were tokenized in units of nouns and adjectives through morphological analysis.

The Twitter library from the konlpy package was used for morphological analysis. Words with only one letter are too short to contain users' opinions, so words only with more than two letters were extracted. Also, domain-specific words were manually added to the morpheme dictionary using the customized_konlpy package. A total of 251 words, including words related to mobile stock trading (e.g., "non-face-to-face," "account opening," "order book," and "trend line") or words related to mobile app usage (e.g., "drag,"

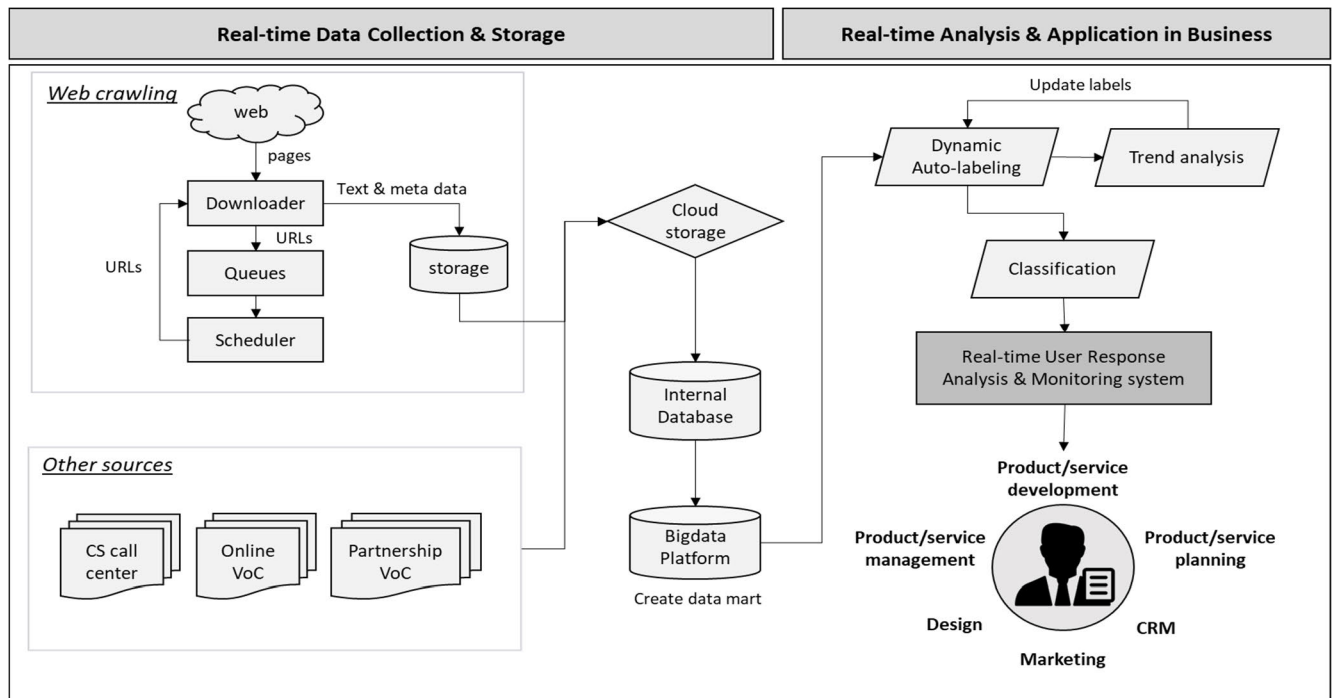


FIGURE 5. Overall process of establishing an automated system.

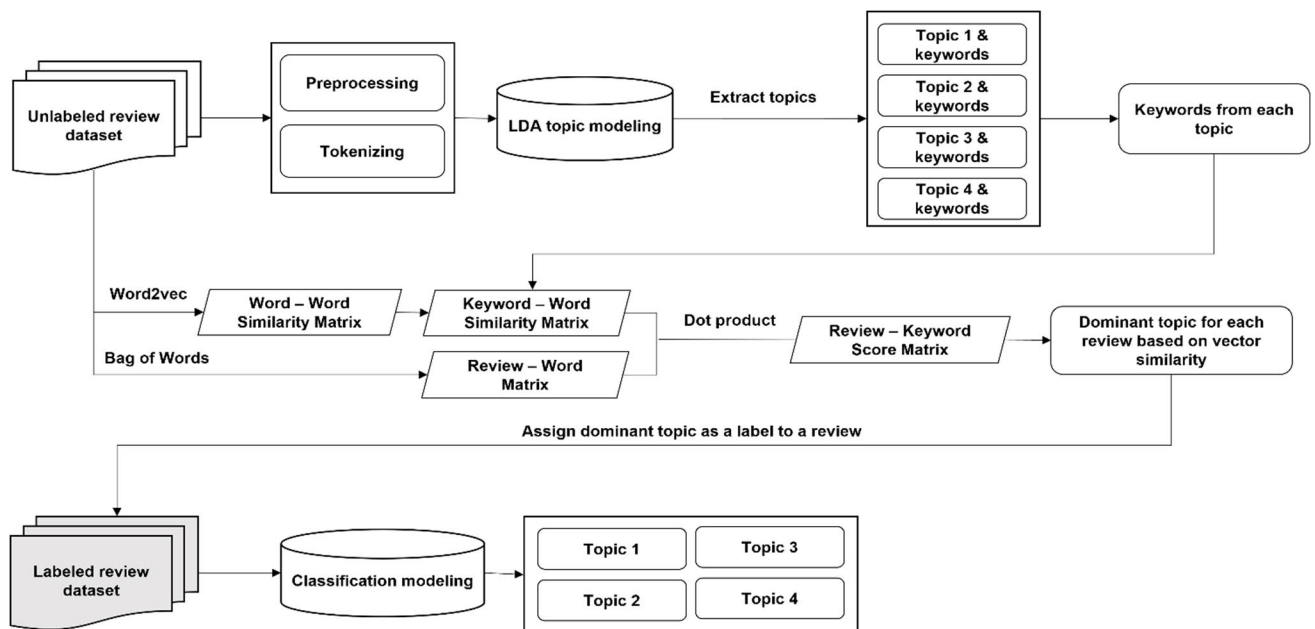


FIGURE 6. Overall process of the proposed method.

“swipe,” and “bounce”) were added to the Twitter morpheme dictionary. Among the extracted words, heteromorphic synonyms were integrated into one word for accurately reflecting the similarity between word vectors in the labeling process. In Table 2, the left column shows some heteromorphic synonyms, and the right column shows the integrated words.

As a result of morphological analysis, a total of 319,211 words was used in the review text, and 10,424 words after excluding duplicates. Each review includes an average of 11 words. Fig. 8 shows the distribution of the number of reviews by length. If the number of extracted words was too small, it would be difficult to determine the relationship between words. Therefore, 28,962 reviews that contain four

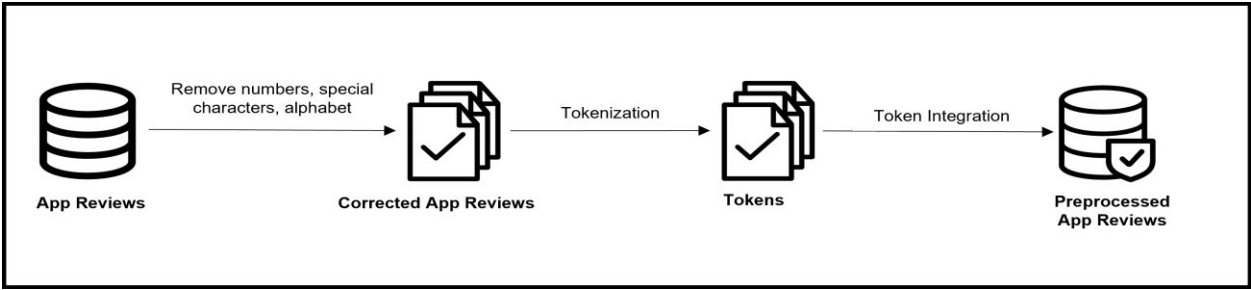


FIGURE 7. Pre-processing text data.

TABLE 1. Previous studies about review classification.

Application Name	Reviews
KB Securities M-able	3,060
KB Securities M-able Mini	590
NH Investment & Securities QV	1,240
SK Securities Jupasu3	1,360
NH Investment & Securities Namu	2,580
Daesin Securities CYBOS	1,470
Daesin Securities Crayon	1,600
Hankook Investment & Securities Ministock	940
Mirae Asset Securities m.ALL	1,650
Mirae Asset Securities m.Stock	2,800
Mirae Asset Securities m.Global	1,130
Samsung Securities mPOP	3,060
Samsung Securities O2	640
Hanwha Investment & Securities STEPS	795
Shinhan Securities Shinhan-i	820
Shinhan Securities Alpha	3,040
Yujin Investment & Securities Smart Champion	1,400
Securities plus	2,960
Kiwoom Securities Hero-S	3,200
Kiwoom Securities Hero-SG	1,580
Hana Investment & Securities Hana OneQ Pro	1,040
Hankook Investment & Securities	2,330
Hankook Investment&Securities efriend Smart	1,160
Hanwha Investment & Securities SmartM	1,260

TABLE 2. Examples of heteromorphic synonyms.

Original tokens	Modified token
Official certificate authentication, joint certificate authentication, etc.	Authentication
Device, (galaxy) flip, (galaxy) note, galaxy, cellphone, etc.	Device
Driver’s license, license, resident registration card, id card, etc.	ID
Excellent, perfect, outstanding, unique, superior, great	Great

or more words were selected as the final dataset for the analysis.

E. TOPIC EXTRACTION

LDA topic modeling is a probability-based model for finding underlying topics in documents. Of the 10,424 words

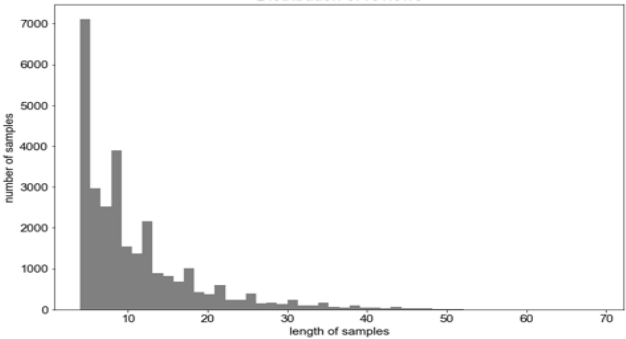


FIGURE 8. Distribution of the data samples by length.

extracted from morphological analysis, a total of 1,080 words, excluding those that appear in more than 50% of documents or less than seven documents, were used for topic modeling. After identifying the perplexity and topic coherence according to the change in the number of topics, the ideal number of topics was determined to be six. Fig. 9 illustrates the change in perplexity and topic coherence according to the change in the number of topics.

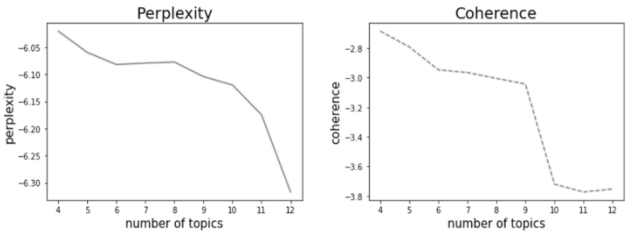


FIGURE 9. Perplexity (left) and topic coherence (right) based on change in number of topics of LDA topic modeling.

Each topic was named by the author given the relationship between the representative words. In addition, five representative words were selected among the top 10 related words for each topic. After converting the top 10 words into vectors using word2vec, the average cosine similarity between a word and the rest of the words was calculated. The top five words with the highest similarity with the rest of the words were selected as representative keywords of the topic.

F. TOPIC LABELING

Fig. 10 demonstrates the topic labeling process using word2vec. Among the 10,424 words extracted from morphological analysis, 1,319 that appeared at least 15 times in the review data were embedded as vectors using the Skip-Gram method. Based on the central word, the number of words to be trained back and forth (which signifies the “window”) was set to two, and each word was converted into a 100-dimensional vector. Next, a Word-Word Similarity Matrix between each word vector was generated using the cosine similarity. A Keyword-Word Similarity Matrix was created by selecting 30 rows that correspond to the representative keywords derived from the topic modeling from Word-Word Similarity Matrix.

In addition, a Review-Word Matrix, which indicates the number of times words appear in each review, was constructed. Each row represented a review, each column consisted of 1,452 words, and the value of the matrix was the number of times that the word appeared in the review. If the Review-Word Matrix consisted of 0 and 1 depending on whether a word appears or not, reviews may not be properly labeled as a dominant topic with higher weights. Therefore, BoW, which is an embedding method based on word frequency, was used to generate the Review-Word Matrix rather than one-hot encoding, which is expressed as 0 and 1.

The Review-Keyword Score Matrix that showed similarity with 30 topic keywords for each review was generated using the inner product of the Keyword-Word Similarity Matrix and the Review-Word Matrix. Afterward, the one having the highest sum of similarity scores was selected as the final label of the review based on the sum of the similarity scores of the representative words of each topic. This is regarded as the correct label and is used for classification modeling.

V. MODEL DESIGN

A. EMBEDDING FOR CLASSIFICATION

For deep learning classification models, the review data were converted to a list of integers for embedding. Unique integers are assigned to the words extracted from the morphological analysis based on word frequency. The 1,300 words that appeared most frequently in the review data (of 10,424 extracted words) were selected to create a word set. If a word did not belong to the word set, it was preserved by encoding it as 1 to form the index of OOV (Out-Of-Vocabulary). As a result, “OOV” is encoded to 1, “Trading” to 2, and “Not working” to 3, with a low integer indicating a word that ranks high, as in Fig. 11. In this study, the tokenizer of TensorFlow was used for integer encoding.

After integer encoding, padding is performed to equalize the review lengths by filling data with specific values. According to Fig. 8, since most reviews can be expressed in 69 words, the length of all reviews was set to 69. In a review less than 69 words, the length was equalized as 69 with the rest of the review data filled with zeros, which is called zero

padding. After padding, an embedding layer from Keras turns the review data into dense vectors of fixed size.

For KoBERT, input text must be converted via token embedding, segment embedding, and position embedding. The token embedding layer extracts the tokens of each review and adds a [CLS] token at the beginning of the review and an [SEP] token at the end of the review. Segment embedding is used to distinguish two sentences and is not necessary for document classification. Therefore, only sentence zero embedding is added. Finally, positional embedding provides information about the position of each word in a sentence. The maximum length of the review data is set to 200, and padding is applied to equalize the length of reviews. Position embedding is generated by calculating the number of actual words from the total length fixed at 200.

B. EXPERIMENTAL SETUP

LSTM, GRU, BiLSTM, CNN, and KoBERT algorithms were used to train classification models after topic labeling. The size of the word sets of an embedding layer of LSTM, GRU, and BiLSTM was set to 1,300, and the embedding dimension was set to 100. The model was composed of two hidden layers of size 128 and 64, and the dropout value was 0.5. The model was trained with a batch size of 32, a learning rate of 0.001, and an Adam optimizer. Epoch was set to 40, but an early stopping criterion was added to end the training process when the validation loss was minimized and unchanged for seven consecutive epochs, a patience score of 7. Last, the SoftMax function is selected for the output layer as well as categorical cross entropy.

For the CNN model, the number of filters in the convolutional layer is set to 256, the filter size is set to 3, and ReLu is used as the activation function. The size of the fully connected layer is 128, the dropout value is 0.5, and the activation function is ReLu. Other hyperparameters are set to be the same as those of the RNN-type model.

KoBERT written in Pytorch was used. The model consists of a single linear layer with a size of 768, and the dropout value was set to 0.5. The model was trained with an Adam optimizer with a batch size of 32, epoch of 10, and learning rate of 0.00005.

C. EVALUATION METRICS

The precision, recall, and F1-score were the model evaluation indicators. Precision refers to the proportion of data that is true among data that the model predicts to be true, and recall refers to the proportion of data predicted to be true among data that are true. The F1-score is an index that combines the means of precision and recall. More accurate predictions result in values closer to 1 for all three indexes. Precision, recall, and F1-score can be expressed as (2) – (4) using elements from the confusion matrix in Table 3.

In multi-class classification, classes (labels) are mutually exclusive, and each review is classified to only one of the multiple classes. Thus, the evaluation metrics are calculated by comparing the true and predicted labels of each instance.

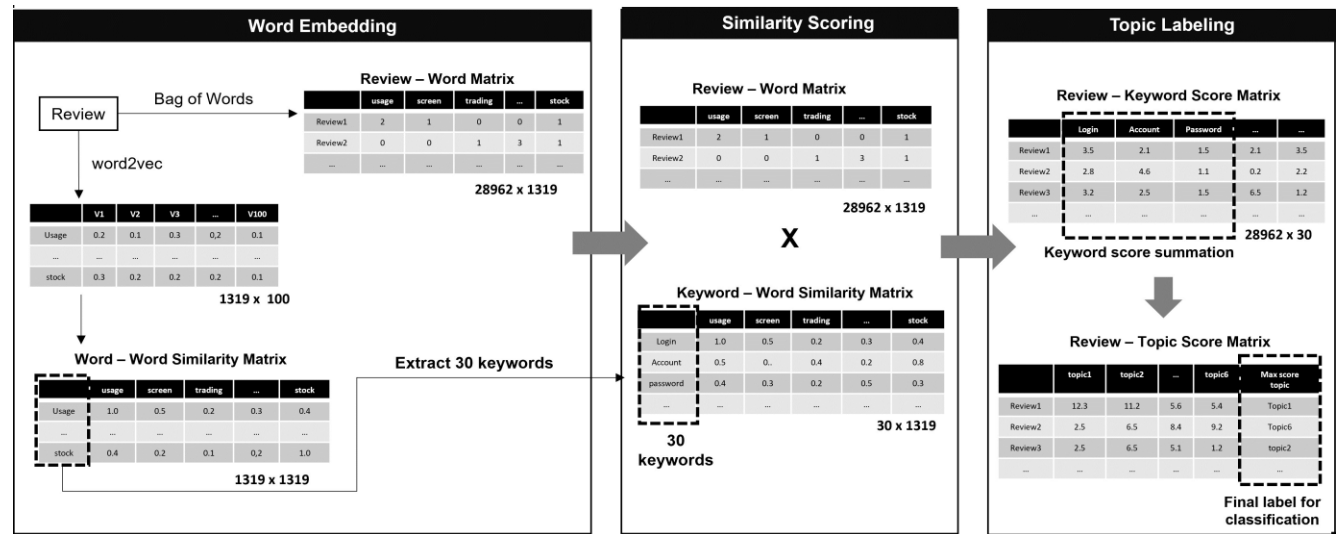


FIGURE 10. Automated labeling process based on word vector similarity.

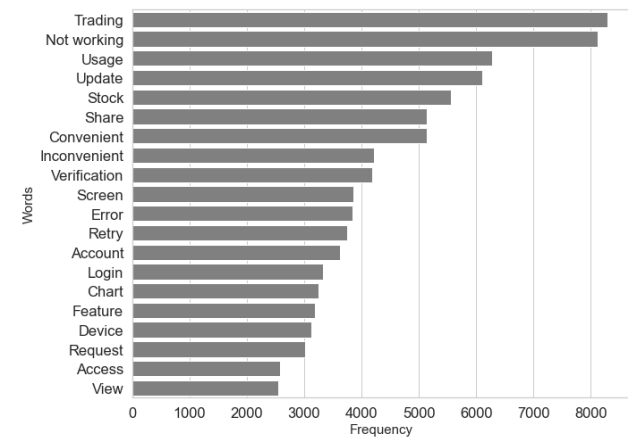


FIGURE 11. Word frequency.

In this study, weighted average was used for evaluation metrics.

$$\text{precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{recall} = \frac{TP}{TP + FN} \quad (3)$$

$$F1 - \text{score} = \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}} \quad (4)$$

TABLE 3. Confusion matrix.

		Predicted	
		Positive	Negative
Actual	Positive	TP	FN
	Negative	FP	TN

VI. EXPERIMENTAL RESULTS

A. KEYWORD ANALYSIS

Fig. 11 shows the frequency ranking of 10,424 words extracted from the morphological analysis. Many words related to app usage (e.g., “usage,” “update,” “login,” and “screen”) as well as words related to stock trading (e.g., “stock,” “trading,” and “account”) appeared. In addition, a number of words, such as “request” and “inconvenient,” showed user needs and feelings when using the app.

Fig. 12 illustrates an association between 10,424 words. The association is measured by “Support,” which refers to the degree to which specific words A and B appear at the same time among all the words in the document. Among all the word pairs, 100 with the highest support value were selected to analyze the association between the words. The darkest-colored nodes of “trading,” “not working,” “update,” and “convenient” were words with higher frequency. Among the words that appeared along with “trading,” words such as “stock,” “not working,” “usage,” and “convenient” had high support. Among the words that appeared along with “not working,” “update,” “login,” “access,” and “retry” showed high support.

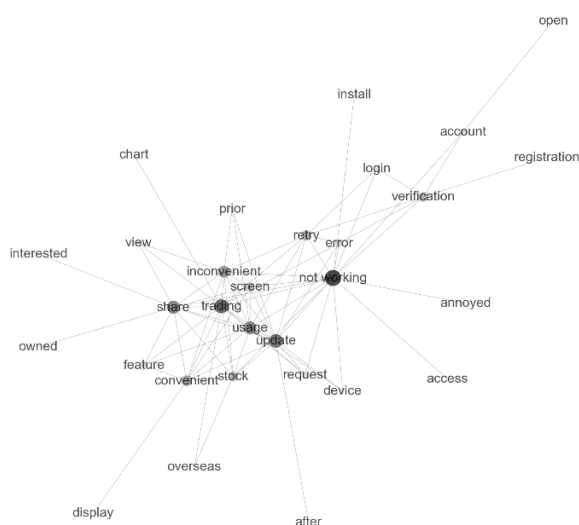
B. TOPIC MODELING

Table 4 shows six topics derived from LDA topic modeling and 10 representative keywords that comprise each topic. However, some words like “convenient” and “usage” appeared as keywords for multiple topics. Thus, it is essential to avoid duplicate content between topics since each review is supposed to be labeled as the most highly related topic based on similarity score.

To resolve this problem, word2vec and document vector representation were applied to calculate the vector similarity and to select the optimal representative keywords. The average value of cosine similarity between each word and the

TABLE 4. Representative keywords for six topics derived from LDA topic modeling.

Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6
Account	Stock	Update	Stock	Login	Chart
Verification	Convenient	Not working	Trading	Trading	Trading
Not working	Usage	Retry	Overseas	Error	Usage
Password	Display	Device	Convenient	Not working	Update
Insert	Interested	Access	Investment	Slow	Inconvenient
Account open	Feature	Error	Use	Bad	Prior
Registration	Menu	Install	Fee	Fast	Feature
Login	Information	Execution	Domestic	Loading	Version
Number	Inconvenient	Issue	Possible	Improvement	Screen
Error	Screen	Disconnection	Widget	Speed	Request

**FIGURE 12.** Network analysis result.

remaining nine words was calculated, and only the top five words with the highest similarity were selected as the final representative keywords of the topic. Table 5 shows the cosine similarity between keywords for each topic.

Table 6 shows the final topics and the selected keywords for each topic. The six topics were “Account open and login,” “Menu usage and user-friendliness,” “Network error,” “Stock trading,” “Speed issue,” and “Usability before and after an update” based on the author’s domain knowledge as well as some official advice from experts in the securities industry. For validation of the topic names, 10 experts (with whom the author worked at an investment & securities firm) were asked to select the most important features of stock trading apps. Details can be found in Appendix A.

The first topic was “Account open and login,” and words such as “verification,” “password,” “login,” “registration,” and “insert” were extracted. When using stock trading apps for the first time, users were required to verify existing accounts or to open a new account to log in to the application.

The second topic was “Menu usage and user-friendliness,” and words such as “convenient,” “menu,” “display,” “inconvenient,” and “interested” were extracted. Stock trading app users should be able to enjoy user-friendly trading-related services such as adding stocks of interest to their watchlist.

The third topic was “Network error,” and words such as “disconnection,” “install,” “execution,” “issue,” and “retry” were extracted. A stable trading system is critical for timing of buying and selling stocks. However, frequently occurring network errors can cause great inconvenience to users.

The fourth topic was “Stock trading,” and words such as “domestic,” “overseas,” “investment,” “stock,” and “convenient” were extracted. Stock trading apps are supposed to provide easy-to-use services for trading stocks and financial products.

The fifth topic was “Speed issue,” and words such as “slow,” “speed,” “loading,” “fast,” and “error” were extracted. Since it is important to trade stocks at the right time, the speed of stock trading apps is more critical than that of other types of mobile apps.

The sixth topic was “Usability before and after an update,” and words such as “prior,” “version,” “inconvenient,” “usage,” and “update” were extracted. Frequent updates can cause inconvenience and give a negative impression to users familiar with the previous version of the apps.

C. LABELING

1) AUTOMATED LABELING

Based on the Review-Keyword Score Matrix that shows the similarity score between the 30 representative keywords for six topics and each review, the topic with the highest

TABLE 5. Vector similarity between the representative keywords for each topic.

Topic	Representative words (similarity between words)
Topic 1	verification(0.677), password(0.670), login(0.655), number(0.655), registration(0.632), insert(0.579), error(0.524), account open(0.524), not working(0.512), account(0.464)
Topic 2	convenient(0.568), menu(0.556), display(0.546), inconvenient(0.514), interested(0.506), use(0.503), feature(0.498), screen(0.493), stock(0.464), information(0.432)
Topic 3	disconnection(0.667), install(0.644), execution(0.627), problem(0.626), retry(0.620), access(0.619), device(0.562), error(0.552), not working(0.535), update(0.461)
Topic 4	domestic(0.547), overseas(0.528), stock(0.503), convenient(0.481), investment(0.481), possible(0.460), trading(0.449), use(0.428), fee(0.408), widget(0.342)
Topic 5	slow(0.616), speed(0.594), loading(0.565), bad(0.496), fast(0.492), error(0.489), login(0.466), not working(0.454), improvement(0.441), trading(0.411)
Topic 6	prior(0.566), version(0.535), inconvenient(0.532), use(0.492), update(0.477), request(0.452), feature(0.451), screen(0.419), chart(0.417), trading(0.374)

TABLE 6. Final representative keywords for each topic.

Topic	Topic description	Representative keywords
Topic 1	Account open and login	Verification, password, login, registration, insert
Topic 2	Menu usage and user-friendliness	Convenient, menu, display, inconvenient, interested
Topic 3	Network error	Disconnection, install, execution, issue, retry
Topic 4	Stock trading	Domestic, overseas, investment, stock, convenient
Topic 5	Speed issue	Slow, speed, loading, fast, error
Topic 6	Usability before and after an update	Prior, version, inconvenient, usage, update

similarity score was assigned to the review as a true label. Table 7 shows the number of reviews assigned to each topic label. Most reviews were labeled as Topic 2, and the fewest reviews were labeled as Topic 5. Despite the imbalance in data size among categories, this dataset was used for training without additional sampling to compare the classification performance between categories.

2) MANUAL LABELING

To verify our proposed auto-labeling method, we performed manual labeling and compared labeling accuracy and time spent on labeling. The accuracy was calculated by counting the number of automatically labeled review topics that matched those of the manually labeled reviews. We also compared approximate time spent on automated and manual labeling.

We first split the entire data into two parts based on the median of the highest topic similarity scores, which turned out to be 18.89. The reason for using the median was to examine the effect of document length on the labeling accuracy, since a high similarity score means that a review is long and consists of many words that can be used for vector similarity calculation. Then, we utilized stratified random sampling to extract 600 samples, 300 from the ‘above-median’ dataset and 300 from the ‘below-median’ dataset, and the 300 samples each consist of 50 samples for each of the six topic labels.

As shown in Table 8, the proposed auto-labeling method showed 77% accuracy with the ‘below-median’ dataset and 85% with the ‘above-median’ dataset, indicating that longer reviews are more likely to show correspondence between automated labeling results and manual labeling results.

In addition, it took us more than two hours to complete the manual labeling of the 600 samples, whereas it took only five minutes to complete the automated annotation of the entire dataset using the proposed framework.

D. MULTI-CLASS CLASSIFICATION

For classification modeling, 28,962 pieces of review data were divided into training and test data at a ratio of 7:3. Since the dataset was imbalanced, stratified random sampling was used to split the data based on the ratio of categories.

Table 9 shows the prediction performance of the classification model for each topic. In the case of Topic 1 through Topic 4, the F1-scores were higher than the other topics. On the other hand, in the case of Topic 5 with a small number of reviews, the F1-scores were relatively low for all the models. Among the five models, RNN-based models such as LSTM, GRU, and BiLSTM showed higher performance, whereas KoBERT showed the lowest. Transformer-based BERT is generally known to have higher-prediction-performance neural networks when used in existing natural language processing tasks, but the model

TABLE 7. Automated labeling results for each topic.

Topic	Topic description	Number of labelled reviews
Topic 1	Account open and login	4,555
Topic 2	Menu usage and user-friendliness	7,417
Topic 3	Network error	7,231
Topic 4	Stock trading	4,344
Topic 5	Speed issue	1,953
Topic 6	Usability before and after an update	3,462
Total		28,962

TABLE 8. Results of manual labeling on 600 samples and labeling accuracy.

Topic	Number of auto-labelling samples	Number of correct manual labelling samples (below median)	Number of correct manual labelling samples (above median)
Topic 1	50	37	43
Topic 2	50	48	47
Topic 3	50	32	34
Topic 4	50	42	44
Topic 5	50	40	47
Topic 6	50	32	40
Total	300	231 (77%)	255 (85%)

does not seem to work well in this study due to the characteristics of the app review dataset. KoBERT is pre-trained on documents that feature refined language, such as Korean Wikipedia, whereas user reviews are often written using unrefined language (e.g., colloquial words, abbreviations, and profanity). For this reason, the KoBERT model seems to have the lowest performance. Overall, LSTM demonstrated the highest performance with 90% precision and an 89% F1-score among all the models as shown in Table 10.

E. DYNAMIC AUTOMATED LABELING

1) CHANGE IN TOPIC LABELS

In this study, stock trading app reviews left from January 2014 to April 2022 were collected. Fig. 13 shows the number of reviews collected by year, and the number of reviews has increased rapidly since 2021. This may be because the number of millennial investors in their 20s and 30s dramatically increased when the country was hit by COVID-19 and the Korean stock market rose to all-time highs during the pandemic.¹

To identify changes in app user response as a result of the rapid change in stock market trends, LDA topic modeling was performed on 9,768 reviews published before 2021 (January 2014 to December 2020) and 19,194 reviews published after 2021 (January 2021 to April 2022). Given the change in perplexity and coherence, as in Fig. 14 and Fig. 15, the

optimal number of topics was 6 before the year 2021 and was 7 after the year 2021.

As shown in Table 11, topics such as “network error,” “stock trading,” “feature request,” “account open,” “usability,” and “ID verification” were derived before 2021. After 2021, as shown in Table 12, the “customer service” topic was derived in addition to the original six topics in Table 11. The appearance of the “customer service” topic possibly resulted from the increase in related inquiries along with the massive influx of individual investors in 2021. The six topics before 2021 were assumed to be identical to those after 2021, except for the “customer service” topic because of its keyword distribution. Most of the keywords for each topic in Table 11 are duplicated in Table 12. For instance, 8 of 10 topic modeling keywords of the “network error” topic before 2021 also appeared after 2021, indicating sufficient similarity of the two sets. In this sense, labels can be periodically updated and used for classification modeling to properly respond to the latest user reactions through dynamic automated labeling.

2) APPLICATION IN BUSINESS

One of the contributions of this study is application of the dynamic automated labeling technique to build an automated system for real-time user response analysis in business. To be specific, the proposed framework can be applied as follows.

- (1) Business experts choose a specific date based on which the topic labels are updated, depending on the business activities or trends of interest.

¹<https://www.busan.com/view/busan/view.php?code=2022091218515110108>

TABLE 9. Classification results by classifiers.

Classifier	Precision	Recall	F1-score
LSTM	0.90	0.89	0.89
GRU	0.89	0.89	0.89
BiLSTM	0.89	0.89	0.89
CNN	0.87	0.87	0.87
KoBERT	0.83	0.83	0.83

TABLE 10. Classification results by classifiers.

Topic (Size)	Metrics	LSTM	GRU	BiLSTM	CNN	KoBERT
Topic 1 (1,365)	Precision	0.95	0.92	0.92	0.89	0.83
	Recall	0.84	0.90	0.91	0.89	0.89
	F1-score	0.89	0.91	0.91	0.89	0.86
Topic 2 (2,225)	Precision	0.94	0.90	0.88	0.88	0.84
	Recall	0.88	0.89	0.92	0.88	0.86
	F1-score	0.91	0.90	0.90	0.88	0.85
Topic 3 (2,169)	Precision	0.87	0.91	0.91	0.90	0.85
	Recall	0.95	0.92	0.92	0.90	0.85
	F1-score	0.91	0.91	0.91	0.90	0.85
Topic 4 (1,303)	Precision	0.90	0.89	0.89	0.85	0.89
	Recall	0.92	0.92	0.90	0.90	0.78
	F1-score	0.91	0.90	0.90	0.88	0.83
Topic 5 (586)	Precision	0.86	0.83	0.83	0.80	0.76
	Recall	0.80	0.85	0.84	0.81	0.61
	F1-score	0.83	0.84	0.84	0.81	0.67
Topic 6 (1,039)	Precision	0.81	0.83	0.89	0.82	0.71
	Recall	0.89	0.80	0.75	0.74	0.79
	F1-score	0.85	0.81	0.81	0.78	0.75

TABLE 11. Topic modeling results of reviews collected before 2021.

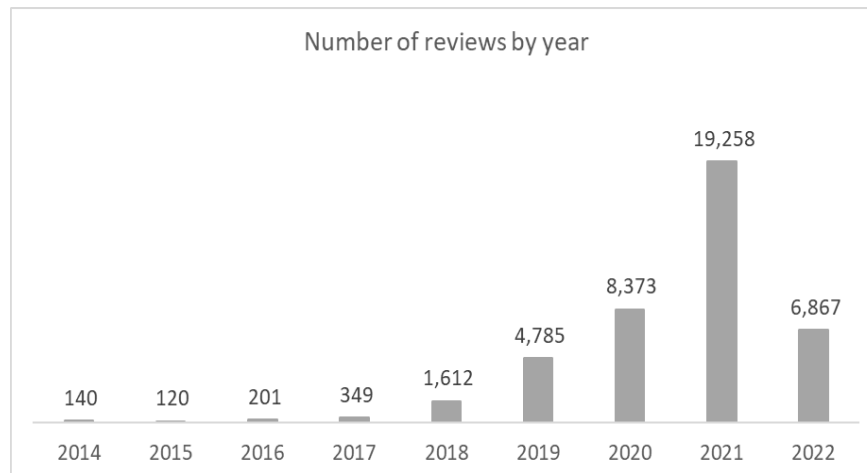
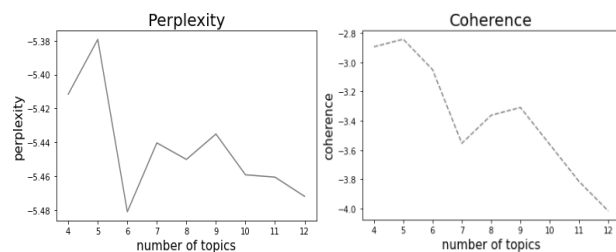
Topic	Topic description	Topic modeling keywords
1	Network error	Not working, update, access, error, retry, login, annoyed, install, loading, execution
2	Stock trading	Trading, stock, overseas, order, balance, owned, account, check, resolve, view
3	Feature request	Share, interested, feature, inconvenient, version, update, addition, view, search, request
4	Account open	Account, usage, device, open, error, slow, setting, fee, notification, event
5	Usability	Convenient, usage, chart, display, screen, information, feature, tidy, inconvenient, prior
6	ID verification	Verification, login, not working, registration, account, password, insert, number, error, simple

- (2) LDA topic modeling is performed on two datasets, one before and one after the chosen date.
- (3) Based on the topic modeling results, it is determined whether the topic labels should be updated or not, given the change in the optimal number

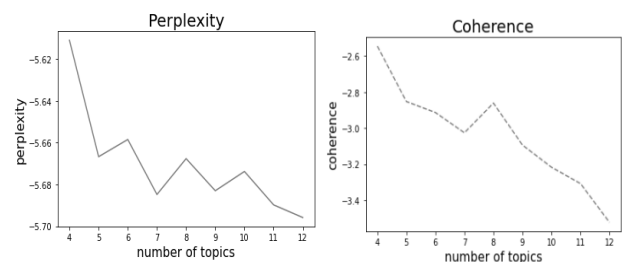
- of topics and the similarity between the derived topics.
- (4) Once the topic labels are updated, business experts can see the results of classifying the latest user responses using visualization dashboards.

TABLE 12. Topic modeling results of reviews collected after 2021.

Topic	Topic description	Topic modeling keywords
1	Account open	Account, open, fee, ID, event, not working, transfer, sign up, pension, check
2	Usability	Convenient, usage, prior, update, display, version, tidy, inconvenient, information, menu
3	Network error	Not working, update, error, access, retry, device, loading, install, issue, execution
4	Customer service	Bad, customer, not working, notification, phone call, customer service, update, manager, trading, connection
5	Feature request	Share, chart, feature, display, inconvenient, interested, setting, request, view, update
6	ID verification	Verification, login, password, insert, not working, registration, account, number, retry, error
7	Stock trading	Trading, stock, overseas, difficult, order, feature, possible, convenient, recommend, domestic

**FIGURE 13.** Number of the collected reviews by year.**FIGURE 14.** Change in perplexity and coherence of reviews collected before 2021.

Since the pace and magnitude of change in business trends and user responses vary by domain, we suggest that business experts can choose a specific date on their own and use it as a standard for topic update. For instance, experts who work for investment & securities firms could pick the date where the

**FIGURE 15.** Change in perplexity and coherence of reviews collected after 2021.

Korean stock market hit record high in 2021 to determine a change in stock trading app user behavior patterns since the chosen date, just like the experiment conducted in this study.

Next, given the LDA topic modeling results, the final topic labels can be updated to cover the latest trends in the text dataset. In this study, the topics derived before and

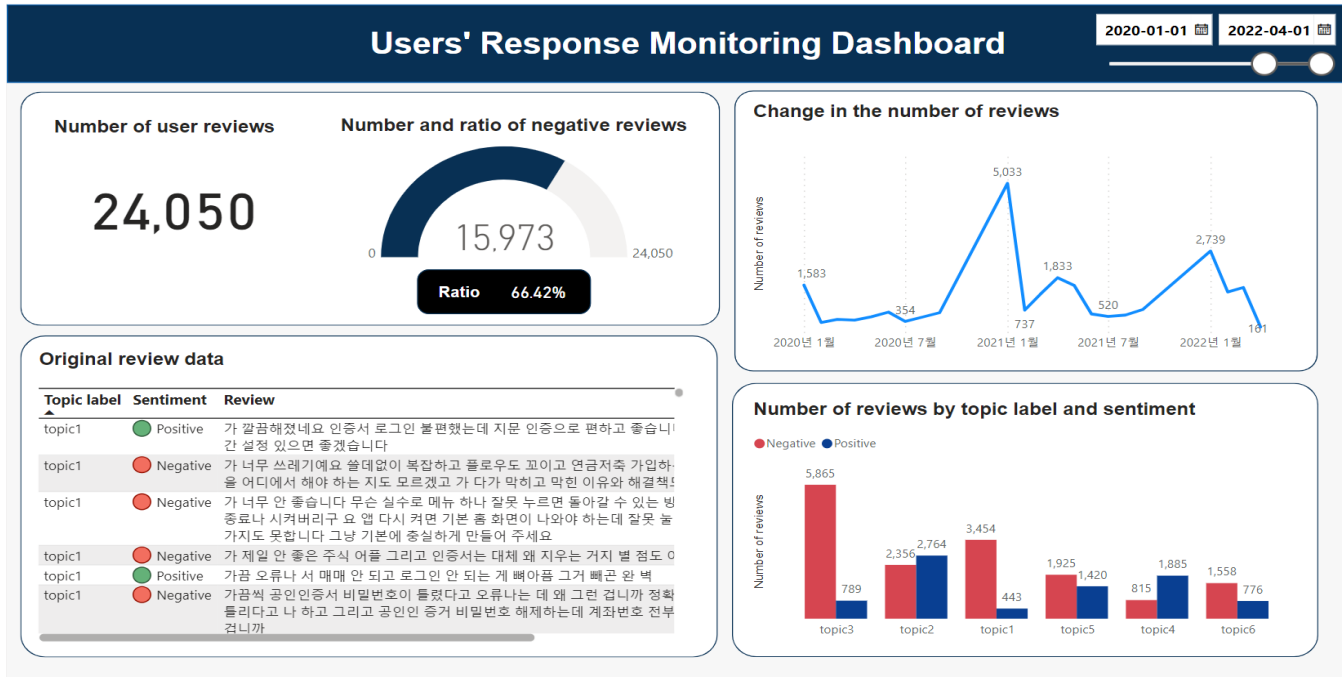


FIGURE 16. A sample visualization dashboard for monitoring users' real-time responses in business.

TABLE 13. Data size of the dataset used in the baseline models.

Dataset	Data size
2012 Sandy Hurricane	9,963
2013 Boston Bombings	9,791
2013 Oklahoma Tornado	9,160
2013 West Texas Explosion	10,006
2013 Alberta Floods	9,049
2013 Queensland Floods	10,016
2011 Joplin Tornado	4,676

after 2021 were sufficiently similar since they consisted of some of the duplicate keywords. To be more precise, the vector similarity between the keywords can be calculated to examine the differences between the topic modeling results and to determine the final updated topic labels. Last, an automated system for real-time user response analysis can be established as a visualization dashboard that business experts can utilize to make informed decisions, as illustrated in Fig. 16.

To support the importance and usability of the automated system in business, various opinions of experts in charge of digital services at a corporation² in which an author was employed were collected. The reviews were collected from

²The author worked at the digital strategy department of an investment & securities firm in Korea. The department is in charge of service planning, development, and operational support of digital channels of the company.

Google Play Store or Appstore, customer voicemails submitted through the company website, a call center hotline log, and user feedback provided by partner companies. A common complaint among experts was manual collection and assessment of user responses. In addition, the experts expressed positive responses about the automated system for real-time user response analysis, and further details can be found in Appendix B.

F. COMPARISON WITH BASELINES APPROACHES

1) BASELINE MODEL AND DATASET

Another experiment was conducted on a disaster-related dataset, representing the social field, to verify the effectiveness of the proposed method on improving the classification performance and generalization regardless of domain. For comparison purposes, the classification results were tested against those of the following baseline models with regard to automatic labeling.

- (1) Xin et al. [24]: Use of named entities for text data
- (2) Alrashdi & O'Keefe [1]: Use of distant supervision
- (3) Si and Aung [23]: Use of expanded vocabulary related to disasters (lexicon)
- (4) Gupta and Joshi [7]: Use of k-means cluster analysis
- (5) Chen et al. [6]: Use of world knowledge (Freebase) and language knowledge (FrameNet)
- (6) Karami et al. [14]: Use of sentiment analysis and topic modeling for situational awareness
- (7) Wahid et al. [22]: Use of Topic Modeling and Topic Ranking Algorithms

TABLE 14. Topic modeling results from the baseline study.

Topic	Representative keywords	Description
Topic 1	Explosion, plant, fertilizer, victim, prayer, tornado, family, affect, injured, news, update, thought	News about injured death and damaged affected by bombing
Topic 2	Texas, people, west, think, break, death, fire, suspect, happen, shit, fuck, camera, blast	People criticism in crisis
Topic 3	Help, rescue, thank, hope, miss, friend, donation, send, hurricane, get	Help seeking
Topic 4	Flood, Queensland, crisis, count, weather, GOD, emergency, north, toll, area, disaster	Update and help seeking in flood crisis
Topic 5	Water, rise, report, house, rain, home, kid, city, job, resident, mayor, evacuate, blame, hurricane	Announcement from government

TABLE 15. Vector similarity between topic modeling keywords of the baseline study.

Topic	Representative words (similarity between words)
Topic 1	Explosion(0.431), plant(0.467), fertilizer(0.470), victim(0.376), prayer(0.467), tornado(0.349), family(0.437), affect(0.448), injured(0.522), news(0.390), update(0.389), thought(0.460)
Topic 2	Texas(0.424), people(0.369), west(0.377), think(0.424), break(0.377), death(0.367), fire(0.372), suspect(0.400), happen(0.438), shit(0.436), fuck(0.428), camera(0.447), blast(0.464)
Topic 3	Help(0.377), rescue(0.318), thank(0.415), hope(0.455), miss(0.397), friend(0.445), donation(0.355), send(0.393), hurricane(0.398), get(0.383)
Topic 4	Flood(0.347), Queensland(0.417), crisis(0.479), count(0.430), weather(0.386), GOD(0.271), emergency(0.408), north(0.396), toll(0.452), area(0.456), disaster(0.470)
Topic 5	Water(0.240), rise(0.318), report(0.305), house(0.342), rain(0.302), home(0.350), kid(0.275), city(0.304), job(0.203), resident(0.407), mayor(0.353), evacuate(0.448), blame(0.259), hurricane(0.241)

In Wahid et al. [22], the classification performance was compared after applying the automatic labeling techniques presented in baseline studies 1 to 6 to a disaster-related Twitter dataset. The present study used that same dataset for comparison purposes with the seven baseline studies.

The dataset used in this study is related to seven disasters that occurred between 2011 and 2013, collected from Imran et al. [12]³ and Olteanu et al. [18].⁴ The seven disasters were 2012 Hurricane Sandy, 2013 Boston Bombings, 2013 Oklahoma Tornado, 2013 West Texas Fertilizer Plant Explosion, 2013 Alberta Floods, 2013 Queensland Floods in Australia, and 2011 Joplin Tornado in Missouri, USA. The dataset consists of Tweet ID, Tweet content in English, publication time, and label. The label column consists of binary labels of related or not to the disaster, except for the Joplin Tornado dataset of which each Tweet was categorized in top-

ics such as ‘Informative,’ ‘Donation,’ ‘Caution and advice,’ and so on. However, automatic labeling was performed with five topics to use the same criteria as Wahid et al. [22] instead of the existing label information. Table 13 shows the size of each dataset.

2) LABELING RESULTS

The collected Twitter dataset also was processed for removal of special characters and numbers, conversion into lower-case letters, and extraction of words of three or more letters through tokenizing. Tokens corresponding to stopwords in nltk were excluded, and lemmatization was performed by converting nouns, adjectives, and verbs into basic forms. As a result, 56,686 Tweets with four or more tokens were selected as the final dataset for analysis. Although Wahid et al. [22] used the original dataset after tokenization, the present study considered the number of extracted tokens to select the final dataset to allow simplicity.

³<http://crisisnlp.qcri.org/>

⁴<http://crisislex.org/data-collections.html#CrisisLexT6>

TABLE 16. Final representative keywords selected based on the vector similarity.

Topic	Representative keywords
Topic 1	Injured, fertilizer, prayer, plant, thought
Topic 2	Blast, camera, happen, shit, fuck
Topic 3	Hope, friend, thank, send, miss
Topic 4	Crisis, disaster, area, toll, count
Topic 5	Evacuate, resident, mayor, home, house

TABLE 17. Comparison of classification results with the baseline approaches.

Baseline	Accuracy	Precision	Recall	F1-score
Xin et al, 2019	82.1%	20.9%	80.8%	81.4%
Alrashdi & O’Keefe, 2020	81.9%	82.7%	81.2%	79.4%
Si et al, 2018	80.1%	82.1%	79.5%	79.8%
Gupta & Joshi, 2021	79.2%	78.2%	78.3%	79.1%
Chen et al, 2017	77.2%	76.4%	75.3%	76.3%
Karami et al, 2020	76.3%	75.8%	76.1%	76.8%
Wahid et al, 2022 (ANN)	80.7%	80.1%	79.4%	79.1%
Wahid et al, 2022 (LSTM)	83.5%	83.1%	82.0%	82.2%
Wahid et al, 2022 (CNN)	83.1%	84.0%	82.0%	83.1%
My approach (LSTM)	90.0%	90.0%	90.0%	90.0%
My approach (CNN)	88.0%	88.0%	88.0%	88.0%

Wahid et al. [22] also used topics derived from LDA topic modeling as true labels for the Twitter dataset. Since the results of probability-based LDA topic modeling can vary depending on the seed, the same topic modeling results from Wahid et al. [22] were applied in the present study to compare model performance under similar conditions. Table 14 shows the topic modeling results presented by Wahid et al. [22].

Based on the results of topic modeling in Table 14, the top five representative words with high vector similarity were selected using the method proposed in this study. Of the final 56,686 Tweets selected for analysis, 3,124 words that appeared more than 15 times were converted into vectors. As shown in Table 15, the cosine similarity between representative words was calculated. The final representative words for each of the five topics can be found in Table 16.

3) CLASSIFICATION RESULTS

LSTM and CNN were used to compare multi-class classification results with the baseline model results. The training

parameters were the same as those in the experiment conducted with the stock trading app review dataset except for the number of final classification classes, which was five in this case. The classification model performance of the baseline approaches and the automatic labeling method proposed in this study are presented in Table 17.

In Wahid et al. [22], the LSTM model showed an accuracy of 83.5%, which was higher than the other six baseline approaches; in the present study, CNN showed a higher accuracy of 88% and LSTM was 90%. For precision and F1-score, the classification performances of the CNN and LSTM models in this study were 88% and 90%, respectively, higher than all the baseline approaches.

Wahid et al. [22] explained the improvement in model performance by assignment of the most context-relevant topic. Similarly, in the present study, topics to be used as true labels were derived through LDA topic modeling, while the representative keywords most related to the topic were selected based on vector similarity and were used to finalize the automatic labeling results. As a result, the labeling method proposed in this study reflects most of the contents

TABLE 18.

What do you think is the most important feature of stock trading apps? (up to three responses allowed)	Number of votes	Ratio
It should provide features for trading and banking	7	23%
App loading should be fast enough to buy and sell stocks at the right time	6	20%
Service errors should be minimized	5	17%
Opening an account should be easy	4	13%
User-friendly interface design is important to find menus	3	10%
Personalized features or contents should be provided	2	7%
Transaction among financial products, such as bonds or funds, should be allowed	2	7%
Entertaining and useful content about investments should be well-displayed to draw user attention and encourage revisit	1	3%
Total	30	100%

TABLE 19.

Position	Purpose of using user review data	Opinions on the automated review analysis system
Product manager mainly in charge of service operation	Identify major issues with the company's app service	It can streamline the work process since the product managers do not need to manually collect and read reviews every day
App developer & graphic designer	Address requests from product managers who identify issues from user reviews	It can help manage work schedule since they can be aware of real-time issues and do not need to handle last-minute requests from product managers
Product managers mainly in charge of partnership service	Identify major issues with partnership services	It enables management of user feedback efficiently since they do not need to manually collect and read scattered users' review data
Product manager mainly in charge of service planning	Discover user complaints and to brainstorm new service ideas based on user feedback	It can expedite the service planning process by focusing on specific features of user interest

of documents, enhancing model performance compared to previous baseline studies.

VII. CONCLUSION AND FUTURE WORK

A. SUMMARY

This study proposed a dynamic automated labeling process for real-time user intention analysis using topic modeling and word2vec. The underlying topics of documents were derived from topic modelling, and the final representative keywords for each topic were chosen based on vector similarity. The final topic labels were determined by calculating similarity

score between the representative keywords and each document. Last, the performance of the classification models (i.e., LSTM, GRU, BiLSTM, CNN, and KoBERT) was analyzed.

The new labeling technique proposed in this study is expected to increase the efficiency of text classification. In previous studies, researchers manually selected categories and labeled documents. To do so, it was necessary to create tagging guidelines and to educate the researchers, and labeling was very expensive and time-consuming. In addition, researchers' error was possible during labeling, leading to unreliable classification results.

On the other hand, in this study, the vector similarity score between the underlying topics and the words used in the document was calculated, and each document was designed labeled as the topic with the highest similarity score. This approach can replace manual labeling and can be applied to different types of text dataset in other industries such as economics, politics, and pharmaceuticals, and so on.

Finally, the labeling technique presented in this study showed higher classification performance than the baseline studies when applied to a disaster-related Twitter dataset. Furthermore, the dynamic automated labeling technique both contributes to the improvement of classification performance and can be used for development of an automated system since rapid changes in user reactions and intentions can be reflected in labeling.

B. LIMITATIONS AND FUTURE RESEARCH

The limitations of this study are as follows. First, text data preprocessing suitable for the characteristics of each language is essential for applying this automatic labeling method. In this study, tokenizing and lemmatization were applied to the English dataset, and morphological analysis and heteromorphic synonym integration were performed for the Korean dataset. Data preprocessing was critical in the present analysis since this labeling method was based on vector similarity. In addition, classification performance may vary depending on tokenization package by language.

A second limitation of this study is that the proposed automated labeling method is based on word2vec. Since word2vec is a word-based embedding technique, the feature representation of text does not include the contextual information of homonyms. For future research, sentence-based embedding techniques such as ELMo could be utilized to capture the contextual meaning of a word sequence in documents.

Last, we conducted a case study of automatic labeling and text classification using two types of datasets due to resource constraints. Research on the end-to-end process for external data collection, database storage, and the establishment of an automated system using an in-house big data platform should be conducted in future studies.

APPENDIX A

See Table 18.

APPENDIX B

See Table 19.

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